

MERCED 23 WSW, CA, USA

WMO: 745070

Lat: 37.2381N

Lon: 120.8825W

Elev: 24

StdP: 101.04

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 93243

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.8	-1.2	-6.1	2.3	8.9	-4.4	2.6	7.7	8.3	11.2	7.1	10.2	0.6	N/A	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	38.2	20.4	36.7	19.9	35.3	19.5	21.8	35.4	21.1	34.4	20.4	33.5	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	11.9	26.2	15.8	11.3	25.3	15.2	10.8	24.5	63.4	35.5	60.6	34.4	58.2	33.5	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.0	5.0	-5.6	40.7	1.6	1.1	-6.8	41.5	-7.7	42.1	-8.6	42.7	-9.8	43.5		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	7.8	10.0	12.4	14.5	17.9	21.6	24.1	23.1	21.0	16.7	11.3	7.5		
	DBStd	6.37	2.63	2.62	2.80	3.19	3.12	3.18	2.64	2.34	3.09	2.79	3.00	3.03		
	HDD10.0	229	77	31	9	2	0	0	0	0	0	0	21	89		
	HDD18.3	1540	326	234	183	121	46	7	0	1	11	67	211	335		
	CDD10.0	2307	9	29	85	138	246	348	436	405	329	207	60	13		
	CDD18.3	577	0	0	1	7	33	106	178	147	91	16	0	0		
	CDH23.3	11048	0	0	43	277	927	2130	3047	2532	1645	433	16	0		
	CDH26.7	6077	0	0	6	92	403	1204	1875	1489	877	132	0	0		
(14) Wind	WSAvg	1.8	1.6	1.9	2.0	2.2	2.2	2.2	2.0	1.8	1.7	1.7	1.4	1.6		
(15) Precipitation	PrecAvg	273	63	57	36	16	15	2	0	0	2	15	23	45		
	PrecMax	547	159	215	113	58	94	7	6	5	13	63	70	91		
	PrecMin	132	7	4	1	0	0	0	0	0	0	0	0	4		
	PrecStd	96	40	44	30	14	21	2	1	1	3	18	18	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	22.2	27.2	32.8	36.3	39.8	40.3	39.0	37.8	33.4	25.3	18.7		
		MCWB	11.7	14.3	16.1	18.9	18.7	21.1	20.8	20.7	19.4	17.8	14.9	11.9		
	2%	DB	16.6	20.3	24.3	29.2	33.6	37.6	38.4	37.2	35.8	29.9	23.1	16.3		
		MCWB	11.4	13.7	15.2	16.7	17.3	20.6	20.3	20.1	19.1	17.1	14.8	11.2		
	5%	DB	15.1	18.3	22.4	26.6	31.1	35.4	36.8	35.5	34.1	27.8	20.8	14.8		
		MCWB	11.1	12.6	14.4	15.6	16.7	19.3	20.1	19.7	18.7	15.8	13.8	10.5		
	10%	DB	13.5	16.6	20.1	23.9	28.7	33.4	35.3	34.0	32.1	25.7	18.6	13.5		
		MCWB	10.7	12.0	13.4	14.7	16.2	18.7	19.9	19.5	18.2	15.3	12.4	9.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	16.7	17.5	20.0	20.3	23.3	22.7	22.2	21.7	19.5	17.4	15.6		
		MCDB	17.3	19.3	24.6	30.0	33.7	37.2	37.0	35.8	33.6	29.5	23.0	17.5		
	2%	WB	13.4	15.3	16.4	17.7	18.6	21.8	21.7	21.3	20.3	18.3	16.0	14.0		
		MCDB	15.5	18.9	22.6	27.3	31.2	35.4	35.7	34.6	32.4	27.9	20.9	15.2		
	5%	WB	12.4	14.3	15.4	16.6	17.6	20.6	20.9	20.5	19.6	17.4	14.6	12.4		
		MCDB	14.4	17.8	20.9	25.4	29.0	33.9	34.8	33.7	31.9	26.0	18.9	13.7		
	10%	WB	11.4	13.1	14.4	15.5	16.6	19.6	20.2	19.9	18.8	16.3	13.5	10.9		
		MCDB	13.4	16.5	19.6	23.1	27.4	32.6	33.8	32.7	31.0	24.0	17.6	12.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.8	14.1	16.7	19.2	21.6	22.0	21.9	20.9	17.5	14.4	11.4		
		MCDBR	14.2	16.2	18.2	21.3	23.2	24.0	22.9	23.1	23.1	21.0	17.5	13.0		
	5% WB	MCWBR	9.2	9.7	9.3	9.3	8.8	8.9	8.5	8.5	8.5	8.9	9.4	9.0		
		MCWBR	9.4	13.8	15.6	19.2	21.3	22.1	21.4	21.5	21.2	18.8	14.6	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.345	0.338	0.357	0.365	0.348	0.363	0.351	0.349	0.362	0.341	0.321		
		taud	2.421	2.427	2.447	2.394	2.385	2.446	2.414	2.426	2.439	2.401	2.438	2.507		
	Ebn at Noon		838	886	928	923	916	927	911	917	902	853	832	842		
		Edh at Noon	87	98	105	117	120	113	116	112	104	99	85	75		
(44) All-Sky Solar Radiation	RadAvg		2.40	3.44	4.91	6.29	7.51	8.35	8.13	7.36	6.07	4.40	2.95	2.18		
	RadStd		0.35	0.39	0.44	0.52	0.39	0.27	0.23	0.19	0.18	0.25	0.17	0.31		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		+0.39	N/A	-1.16	N/A	N/A	N/A	N/A	-112	+140	+35	

Nomenclature: See separate page